August 16, 2016

Ms. Amy Vail
City of Cadillac WWTP
200 N. Lake St.
Cadillac, MI 49601

Subject: Executive Summary for Project Number 210226

Dear Ms. Vail:

Please find the attached biomonitoring report for testing conducted August 04, 2016 through August 11, 2016. The results for this evaluation are:

Test Species	TUa	TUc	MATC
Ceriodaphnia dubia	0	0	>100
Fathead Minnows	0	0	>100

The exposure tests were valid since they met the criteria for survival, reproduction, and growth in the controls. Should you have any questions pertaining to this report please let me know.

Sincerely,
Michael Amboyan
Paragon Laboratories, Inc.

DEQ

MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY - WATER BUREAU

CERIODAPHNIA DUBIA CHRONIC TOXICITY TEST REPORT

By authority of PA 451 of 1994, as amended.

1. NAME OF FACILITY (on NPDES permit)	2. NPDES PERMIT #
Cadillac WWTP	MI0020257

3. RECEIVING WATER (as designated in permit)	4. OUTFALL	5. RECEIVING WATER CONCENTRATION (if known)
Saline River	001	Unknown-Cerio + FH

6. TEST LAB (Name and Address)
Paragon Laboratories, Inc. 12649 Richfield Ct. Livonia MI 48150

7. TEST START DATE	8. TEST END DATE	9. AGE RANGE OF ORGANISMS AT TEST START	10. Report Date
08/04/16	08/10/16	Less Than 24 Hours	8/16/2016 8:37:57 AM

11. NAME OF PERSON CONDUCTING TEST	12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT
Michael Amboyan	Paragon Laboratories, Inc. (734) 462-3900

13. SAMPLE COLLECTION DATES	14. DATE RECEIVED	15. DATE OF FIRST USE	16. ARRIVAL TEMP (°C)
8/3/2016 7:00:00 AM	8/4/2016 10:20:00 AM	08/04/2016	Sample 1: 5 on ice
8/4/2016 7:10:00 AM	8/5/2016 10:05:00 AM	08/06/2016	Sample 2: 4 on ice
8/8/2016 7:00:00 AM	8/9/2016 10:30:00 AM	08/09/2016	Sample 3: 5 on ice

17. TOTAL RESIDUAL CHLORINE	18. AMMONIA (mg/L as N)	19. pH	20. Alkalinity	21. Hardness
Sample 1: <0.02	Sample 1: <0.5	Sample 1: 7.18	Sample 1: 98.3	Sample 1: 201.0
Sample 2: <0.02	Sample 2: <0.5	Sample 2: 7.30	Sample 2: 122.2	Sample 2: 220.8
Sample 3: <0.02	Sample 3: <0.5	Sample 3: 7.48	Sample 3: 94.9	Sample 3: 192.4

22. WAS SAMPLE DECHLORINATED?	23. DESCRIBE DECHLORINATION (if any)
Sample 1: No Sample 2: No Sample 3: No	N/A

24. EFFLUENT SAMPLES WERE COLLECTED:
Facility Does Not Chlorinate

25. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded hold time.)
N/A

26. EFFLUENT FILTERED?	27. STATE MESH SIZE OF FILTER (if filtered)
No	N/A

28. EFFLUENT SAMPLE TYPE:	29. IDENTIFY THE DILUENT (O1)
Sample 1: 24-Hour Composite	MHW
Sample 2: 24-Hour Composite	IDENTIFY THE SECONDARY (O2)
Sample 3: 24-Hour Composite	N/A

30. SUMMARY OF DATA AND RESULTS - SURVIVAL AND REPRODUCITON

CONCENTRATION OF EFFLUENT (%)	O1	5	15	30	60	100	
48-HOUR SURVIVAL (%)	100	100	100	100	100	100	
6-Day MEAN REPRODUCTION/FEMALE	24.5	21.5	21.8	24.9	24.3	24.4	
6-Day MEAN SURVIVAL (%)	100.0	100.0	100.0	100.0	100.0	100.0	

31. 48-HOUR LC50 (%)	32. TUa (acute toxic units)
>100	0

33. 6-Day CHRONIC VALUE (%)	34. NOEC	35. LOEC	36. TUc (chronic toxic units)
>100	100	>100	0

Prepared by Paragon Laboratories, Inc. Livonia MI

DEQ

MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY - WATER BUREAU

FATHEAD MINNOW CHRONIC TOXICITY TEST REPORT

By authority of PA 451 of 1994, as amended.

1. NAME OF FACILITY (on NPDES permit) Cadillac WWTP	2. NPDES PERMIT # MI0020257

3. RECEIVING WATER (as designated in permit) Saline River	4. OUTFALL 001	5. RECEIVING WATER CONCENTRATION (if known) Unknown-Cerio + FH

6. TEST LAB (Name and Address)
Paragon Laboratories, Inc. 12649 Richfield Ct. Livonia MI 48150

7. TEST START DATE 08/04/16	8. TEST END DATE 08/11/16	9. AGE RANGE OF ORGANISMS AT TEST START Less Than 24 Hours	10. Report Date 8/16/2016 8:37:57 AM

11. NAME OF PERSON CONDUCTING TEST Michael Amboyan	12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT Paragon Laboratories, Inc. (734) 462-3900

13. SAMPLE COLLECTION DATES	14. DATE RECEIVED	15. DATE OF FIRST USE	16. ARRIVAL TEMP (°C)
8/3/2016 7:00:00 AM	8/4/2016 10:20:00 AM	08/04/2016	Sample 1: 5 on ice
8/4/2016 7:10:00 AM	8/5/2016 10:05:00 AM	08/06/2016	Sample 2: 4 on ice
8/8/2016 7:00:00 AM	8/9/2016 10:30:00 AM	08/09/2016	Sample 3: 5 on ice

17. TOTAL RESIDUAL CHLORINE	18. AMMONIA (mg/L as N)	19. pH	20. Alkalinity	21. Hardness
Sample 1: <0.02	Sample 1: <0.5	Sample 1: 7.18	Sample 1: 98.3	Sample 1: 201.0
Sample 2: <0.02	Sample 2: <0.5	Sample 2: 7.30	Sample 2: 122.2	Sample 2: 220.8
Sample 3: <0.02	Sample 3: <0.5	Sample 3: 7.48	Sample 3: 94.9	Sample 3: 192.4

22. WAS SAMPLE DECHLORINATED? Sample 1: No Sample 2: No Sample 3: No	23. DESCRIBE DECHLORINATION (if any) N/A

24. EFFLUENT SAMPLES WERE COLLECTED:
Facility Does Not Chlorinate

25. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded hold time.)
N/A

26. EFFLUENT FILTERED? No	27. STATE MESH SIZE OF FILTER (if filtered) N/A

28. EFFLUENT SAMPLE TYPE: Sample 1: 24-Hour Composite Sample 2: 24-Hour Composite Sample 3: 24-Hour Composite	29. IDENTIFY THE DILUENT (O1) MHW IDENTIFY THE SECONDARY (O2) N/A

30. SUMMARY OF DATA AND RESULTS - SURVIVAL AND REPRODUCITON

CONCENTRATION OF EFFLUENT (%)	O1	5	15	30	60	100	
96-HOUR SURVIVAL (%)	100	90	100	95	97.5	90	
7-DAY MEAN BIOMASS (mg/initial fish)	0.2973	0.3123	0.3065	0.3070	0.3058	0.3843	
7-DAY MEAN SURVIVAL (%)	92.5	90	97.5	92.5	92.5	90	

31. 96-HOUR LC50 (%) >100	32. TUa (acute toxic units) 0

33. 7-DAY CHRONIC VALUE (%) >100	34. NOEC 100	35. LOEC >100	36. TUc (chronic toxic units) 0

Prepared by Paragon Laboratories, Inc. Livonia MI

TOXICITY EVALUATION
Ceriodaphnia Dubia Summary Table

Client: City of Cadillac WWTP

Project Number: 210226

Primary Analyst: Michael Amboyan

Effluent Concentration	TOTAL LIVE YOUNG IN FIRST 3 BROODS PER REPLICATE										Total No. of Young	Total No. of Adults
	Replicate Number											
	1	2	3	4	5	6	7	8	9	10		
0%	31	30	27	22	15	33	30	11	12	34	245	10
5%	29	30	31	16	32	15	29	12	8	13	215	10
15%	14	28	33	9	35	13	30	11	15	30	218	10
30%	32	33	14	33	11	11	26	27	30	32	249	10
60%	16	29	50	11	8	12	34	30	29	24	243	10
100%	28	16	15	12	32	30	37	9	32	33	244	10

Qualifier Key:

A = Natural Death

B = Abnormal appearance of organism.

C = Male.

D = No developing young visible in brood pouch throughout test.

E = Aborted young.

F = Young produced but dead at end of the 24 hour period.

G = Young produced in the brood pouch but not released

H = Transfer death.

TOXICITY EVALUATION
Fathead Minnow Summary Table

Client: City of Cadillac WWTP

Project Number: 210226

Primary Analyst: Michael Amboyan

Conc.	Rep. No.	Day 7 Survivability	Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
0%	1	9	0.40696	0.40951	2.55	0.26
		A				
	2	9	0.41604	0.41896	2.92	0.29
		A				
	3	9	0.4092	0.41213	2.93	0.29
		A				
	4	10	0.41045	0.41394	3.49	0.35
5%	5	9	0.41326	0.41649	3.23	0.32
		A				
	6	9	0.40627	0.4094	3.13	0.31
		A				
	7	10	0.4143	0.41728	2.98	0.3
	8	8	0.41296	0.41611	3.15	0.32
		A				
15%	9	10	0.39877	0.40191	3.14	0.31
	10	10	0.41345	0.41632	2.87	0.29
	11	9	0.40708	0.40973	2.65	0.26
		A				
	12	10	0.41047	0.41407	3.6	0.36

Qualifier Key:

A = Natural Death
B = Fungal filaments present
C = Lethargy
D = Excessive ventilation
E = Erratic or uncoordinated swimming
F = Transfer death

TOXICITY EVALUATION
Fathead Minnow Summary Table

Client: City of Cadillac WWTP

Project Number: 210226

Primary Analyst: Michael Amboyan

30%	13	10	0.40618	0.40895	2.77	0.28
	14	8	0.39855	0.40162	3.07	0.31
		A				
	15	9	0.41278	0.41591	3.13	0.31
		A				
	16	10	0.41871	0.42202	3.31	0.33
60%	17	8	0.41165	0.41463	2.98	0.3
		A				
	18	10	0.41375	0.41615	2.4	0.24
	19	10	0.40645	0.40985	3.4	0.34
	20	9	0.40982	0.41327	3.45	0.34
		A				
100%	21	9	0.40534	0.41156	6.22	0.62
		A				
	22	9	0.41493	0.41807	3.14	0.31
		A				
	23	10	0.41211	0.415	2.89	0.29
	24	8	0.40528	0.4084	3.12	0.31
		A				

Qualifier Key:

A = Natural Death

B = Fungal filaments present

C = Lethargy

D = Excessive ventilation

E = Erratic or uncoordinated swimming

F = Transfer death

TOXICITY EVALUATION
Ceriodaphnia Survival and Reproduction Test

Client: City of Cadillac WWTP

Project Number: 210226

Primary Analyst: Michael Amboyan

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		1	2	3	4	5	6	7	8	9	10			
0%	0*	1	1	1	1	1	1	1	1	1	1	0	10	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	4	4	5	0	4	5	5	5	4	4	40	10	4.0
	4	9	10	7	9	11	10	7	6	8	8	85	10	8.5
	5	18	16	15	13	0	0	0	0	0	0	62	10	6.2
	6	0	0	0	0	0	18	18	0	0	22	58	10	5.8
	7													
	8													
	Total	31	30	27	22	15	33	30	11	12	34	245	10	24.5

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		11	12	13	14	15	16	17	18	19	20			
5%	0*	1	1	1	1	1	1	1	1	1	1	0	10	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	4	4	4	5	2	3	3	4	3	4	36	10	3.6
	4	10	12	12	11	10	12	6	8	5	9	95	10	9.5
	5	15	14	15	0	0	0	0	0	0	0	44	10	4.4
	6	0	0	0	0	20	0	20	0	0	0	40	10	4.0
	7													
	8													
	Total	29	30	31	16	32	15	29	12	8	13	215	10	21.5

*Neonates less than 24 hours old added to each replicate vessel at start of test. Values are not included in replicate total.

TOXICITY EVALUATION
Ceriodaphnia Survival and Reproduction Test

Client: City of Cadillac WWTP

Project Number: 210226

Primary Analyst: Michael Amboyan

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		21	22	23	24	25	26	27	28	29	30			
15%	0*	1	1	1	1	1	1	1	1	1	1	0	10	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	6	5	3	4	5	4	4	4	4	4	43	10	4.3
	4	8	8	10	5	10	9	8	7	11	11	87	10	8.7
	5	0	15	0	0	0	0	0	0	0	0	15	10	1.5
	6	0	0	20	0	20	0	18	0	0	15	73	10	7.3
	7													
	8													
	Total	14	28	33	9	35	13	30	11	15	30	218	10	21.8

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		31	32	33	34	35	36	37	38	39	40			
30%	0*	1	1	1	1	1	1	1	1	1	1	0	10	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	4	4	4	5	3	3	5	5	5	4	42	10	4.2
	4	10	12	10	8	8	8	0	10	8	8	82	10	8.2
	5	18	17	0	0	0	0	0	12	17	0	64	10	6.4
	6	0	0	0	20	0	0	21	0	0	20	61	10	6.1
	7													
	8													
	Total	32	33	14	33	11	11	26	27	30	32	249	10	24.9

*Neonates less than 24 hours old added to each replicate vessel at start of test. Values are not included in replicate total.

TOXICITY EVALUATION
Ceriodaphnia Survival and Reproduction Test

Client: City of Cadillac WWTP

Project Number: 210226

Primary Analyst: Michael Amboyan

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		41	42	43	44	45	46	47	48	49	50			
60%	0*	1	1	1	1	1	1	1	1	1	1	0	10	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	4	4	5	4	5	4	4	5	6	3	44	10	4.4
	4	12	9	9	7	3	8	8	4	6	5	71	10	7.1
	5	0	0	18	0	0	0	0	0	17	16	51	10	5.1
	6	0	16	0	0	0	0	22	21	0	0	77	10	7.7
	7													
	8													
	Total	16	29	50	11	8	12	34	30	29	24	243	10	24.3

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		51	52	53	54	55	56	57	58	59	60			
100%	0*	1	1	1	1	1	1	1	1	1	1	0	10	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	5	4	5	5	5	5	5	4	4	4	46	10	4.6
	4	11	12	10	7	9	8	10	5	7	8	87	10	8.7
	5	0	0	0	0	18	17	0	0	0	0	35	10	3.5
	6	12	0	0	0	0	0	22	0	21	21	76	10	7.6
	7													
	8													
	Total	28	16	15	12	32	30	37	9	32	33	244	10	24.4

*Neonates less than 24 hours old added to each replicate vessel at start of test. Values are not included in replicate total.

TOXICITY EVALUATION
Fathead Minnow
Larval Survival and Growth Test

Client: City of Cadillac WWTP

Project Number: 210226

Primary Analyst: Michael Amboyan

Conc.	Rep. No.	Day								Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
		0	1	2	3	4	5	6	7				
0%	1	10	10	10	10	10	10	9	9	0.40696	0.40951	2.55	0.26
	2	10	10	10	10	10	10	9	9	0.41604	0.41896	2.92	0.29
	3	10	10	10	10	10	10	10	9	0.4092	0.41213	2.93	0.29
	4	10	10	10	10	10	10	10	10	0.41045	0.41394	3.49	0.35

Conc.	Rep. No.	Day								Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
		0	1	2	3	4	5	6	7				
5%	5	10	9	9	9	9	9	9	9	0.41326	0.41649	3.23	0.32
	6	10	10	9	9	9	9	9	9	0.40627	0.4094	3.13	0.31
	7	10	10	10	10	10	10	10	10	0.4143	0.41728	2.98	0.3
	8	10	8	8	8	8	8	8	8	0.41296	0.41611	3.15	0.32

Conc.	Rep. No.	Day								Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
		0	1	2	3	4	5	6	7				
15%	9	10	10	10	10	10	10	10	10	0.39877	0.40191	3.14	0.31
	10	10	10	10	10	10	10	10	10	0.41345	0.41632	2.87	0.29
	11	10	10	10	10	10	10	9	9	0.40708	0.40973	2.65	0.26
	12	10	10	10	10	10	10	10	10	0.41047	0.41407	3.6	0.36

TOXICITY EVALUATION
Fathead Minnow
Larval Survival and Growth Test

Client: City of Cadillac WWTP

Project Number: 210226

Primary Analyst: Michael Amboyan

Conc.	Rep. No.	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
30%	13	10	10	10	10	10	10	10	10	0.40618	0.40895	2.77	0.28
	14	10	8	8	8	8	8	8	8	0.39855	0.40162	3.07	0.31
	15	10	10	10	10	10	10	10	9	0.41278	0.41591	3.13	0.31
	16	10	10	10	10	10	10	10	10	0.41871	0.42202	3.31	0.33

Conc.	Rep. No.	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
60%	17	10	9	9	9	9	9	8	8	0.41165	0.41463	2.98	0.3
	18	10	10	10	10	10	10	10	10	0.41375	0.41615	2.4	0.24
	19	10	10	10	10	10	10	10	10	0.40645	0.40985	3.4	0.34
	20	10	10	10	10	10	10	10	9	0.40982	0.41327	3.45	0.34

Conc.	Rep. No.	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
100%	21	10	10	10	10	9	9	9	9	0.40534	0.41156	6.22	0.62
	22	10	10	10	10	9	9	9	9	0.41493	0.41807	3.14	0.31
	23	10	10	10	10	10	10	10	10	0.41211	0.415	2.89	0.29
	24	10	10	10	10	8	8	8	8	0.40528	0.4084	3.12	0.31

TOXICITY EVALUATION
Ceriodaphnia Dubia and Fathead Minnow
Routine Chemical and Physical Determination

Client: City of Cadillac WWTP

Project Number: 210226

Primary Analyst: Michael Amboyan

Parameter	Units	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
0% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	8.2	8.2	8.2	8.6	8.2	8.6	8.6	
pH	S.U.	8.0	8.0	8.0	8.0	8.0	8.0	8.0	
Conductivity	umhos/cm	300	290	310	290	290	310	320	
0% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	7.7	8.2	9.1	8.3	9.3	8.8	0.0	
DO Fathead	mg/L	7.2	7.6	8.0	7.5	7.8	7.7	6.8	
pH Cerio	S.U.	7.9	7.9	8.0	8.0	8.0	8.0	0.0	
pH Fathead	S.U.	8.0	7.7	8.0	7.7	8.0	8.0	8.0	
Conductivity Cerio	umhos/cm	320	320	320	320	310	320	0.0	
Conductivity Fathead	umhos/cm	310	310	320	310	320	320	300	
5% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	8.3	8.2	8.4	9.0	8.4	8.6	8.5	
pH	S.U.	7.6	7.8	7.6	7.6	7.7	7.9	7.6	
Conductivity	umhos/cm	360	390	350	360	350	390	340	
5% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	7.6	8.2	9.1	8.3	9.2	8.7	0.0	
DO Fathead	mg/L	7.1	7.4	8.0	7.3	7.5	8.0	6.9	
pH Cerio	S.U.	7.5	7.8	7.9	7.9	7.9	7.8	0.0	
pH Fathead	S.U.	6.9	7.6	7.6	7.5	7.6	8.0	7.8	
Conductivity Cerio	umhos/cm	390	430	360	400	380	420	0.0	
Conductivity Fathead	umhos/cm	400	400	360	380	360	390	410	

TOXICITY EVALUATION
Ceriodaphnia Dubia and Fathead Minnow
Routine Chemical and Physical Determination

Client: City of Cadillac WWTP

Project Number: 210226

Primary Analyst: Michael Amboyan

Parameter	Units	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
15% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	8.4	8.3	8.5	9.0	8.4	8.5	8.7	
pH	S.U.	7.3	7.6	7.6	7.6	7.5	7.8	7.4	
Conductivity	umhos/cm	420	470	440	430	470	460	450	
15% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	7.5	8.1	9.1	8.4	9.2	8.6	0.0	
DO Fathead	mg/L	7.2	7.4	7.8	7.2	7.3	7.5	7.0	
pH Cerio	S.U.	7.5	7.8	7.9	7.9	7.8	8.0	0.0	
pH Fathead	S.U.	6.8	7.6	7.6	7.5	7.5	7.9	7.6	
Conductivity Cerio	umhos/cm	450	520	450	470	500	520	0.0	
Conductivity Fathead	umhos/cm	460	450	440	450	500	480	470	
30% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	8.5	8.3	8.5	8.9	8.3	8.3	8.6	
pH	S.U.	7.2	7.4	7.5	7.5	7.5	7.8	7.4	
Conductivity	umhos/cm	530	610	550	570	610	570	550	
30% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	7.5	8.1	9.2	8.5	9.1	8.7	0.0	
DO Fathead	mg/L	7.2	7.4	7.9	7.2	7.1	7.5	6.9	
pH Cerio	S.U.	7.6	7.7	7.8	7.8	7.7	7.9	0.0	
pH Fathead	S.U.	6.9	7.5	7.6	7.4	7.4	7.4	7.6	
Conductivity Cerio	umhos/cm	580	580	590	630	670	620	0.0	
Conductivity Fathead	umhos/cm	580	600	560	590	630	580	580	

TOXICITY EVALUATION
Ceriodaphnia Dubia and Fathead Minnow
Routine Chemical and Physical Determination

Client: City of Cadillac WWTP

Project Number: 210226

Primary Analyst: Michael Amboyan

Parameter	Units	Day							
		0	1	2	3	4	5	6	7

60% - Initial

Parameter	Units	0	1	2	3	4	5	6	7
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	8.6	8.4	8.3	8.8	8.1	8.1	8.6	
pH	S.U.	7.2	7.4	7.4	7.4	7.5	7.5	7.4	
Conductivity	umhos/cm	790	830	840	850	840	860	830	

60% - Final

Parameter	Units	0	1	2	3	4	5	6	7
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	7.4	8.1	8.8	8.5	9.4	8.7	0.0	
DO Fathead	mg/L	7.2	7.4	7.7	7.1	7.1	7.5	6.8	
pH Cerio	S.U.	7.5	7.8	7.6	7.6	7.6	793.0	0.0	
pH Fathead	S.U.	6.9	7.6	7.5	7.5	7.4	7.4	7.3	
Conductivity Cerio	umhos/cm	830	830	880	900	910	920	0.0	
Conductivity Fathead	umhos/cm	850	820	860	870	880	870	870	

100% - Initial

Parameter	Units	0	1	2	3	4	5	6	7
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	9.0	8.4	8.2	8.6	8.0	7.9	8.5	
pH	S.U.	7.2	7.2	7.4	7.5	7.5	7.4	7.4	
Conductivity	umhos/cm	1200	1200	1200	1200	1200	1200	1200	

100% - Final

Parameter	Units	0	1	2	3	4	5	6	7
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	7.5	8.2	8.9	8.4	8.7	8.6	0.0	
DO Fathead	mg/L	7.3	7.3	7.6	6.9	7.1	7.4	7.0	
pH Cerio	S.U.	7.6	7.9	7.6	7.5	7.6	7.6	0.0	
pH Fathead	S.U.	6.9	7.5	7.6	7.5	7.5	7.4	7.2	
Conductivity Cerio	umhos/cm	1200	1200	1200	1200	1300	1300	0.0	
Conductivity Fathead	umhos/cm	1200	1200	1200	1200	1200	1200	1200	

Ceriodaphnia Reproduction 210226

Title: Ceriodaphnia Reproduction City of Cadillac WWTP 210226
File: cr210226.txt Transform: NO TRANSFORMATION

Kolmogorov Test for Normality

D = 0.1845 (p-value = 0.0000)
D* = 1.4472

Critical D* = 1.035 (alpha = 0.01 , N = 60)
= 0.895 (alpha = 0.05 , N = 60)

Data FAIL normality test (alpha = 0.01). Try another transformation.

Warning - The first three homogeneity tests are sensitive to non-normality and should not be performed with this data as is.

Bartlett's Test for Homogeneity of Variance

Calculated B1 statistic = 1.6735 (p-value = 0.8922)

Data PASS B1 homogeneity test at 0.01 level. Continue analysis.

Critical B = 15.0863 (alpha = 0.01, df = 5)
= 11.0705 (alpha = 0.05, df = 5)

Summary Statistics on Data TABLE 1 of 2

GRP	IDENTIFICATION	N	MIN	MAX	MEAN
1	Control	10	11.0000	34.0000	24.5000
2	5%	10	8.0000	32.0000	21.5000
3	15%	10	9.0000	35.0000	21.8000
4	30%	10	11.0000	33.0000	24.9000
5	60%	10	8.0000	50.0000	24.3000
6	100%	10	9.0000	37.0000	24.4000

Summary Statistics on Data TABLE 2 of 2

GRP	IDENTIFICATION	VARIANCE	SD	SEM	C.V. %
1	Control	78.5000	8.8600	2.8018	36.1634
2	5%	89.1667	9.4428	2.9861	43.9200
3	15%	104.1778	10.2068	3.2277	46.8200
4	30%	85.4333	9.2430	2.9229	37.1205
5	60%	166.0111	12.8845	4.0744	53.0228
6	100%	104.7111	10.2328	3.2359	41.9379

ANOVA Table

SOURCE	DF	SS	MS	F
Between	5	112.7333	22.5467	0.2154
Within (Error)	54	5652.0000	104.6667	
Total	59	5764.7333		

(p-value = 0.9545)

Critical F = 3.3769 (alpha = 0.01, df = 5,54)
= 2.3861 (alpha = 0.05, df = 5,54)

Since F < Critical F FAIL TO REJECT Ho: All equal (alpha = 0.01)

Dunnett's Test - TABLE 1 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	TRANSFORMED MEAN	MEAN CALCULATED IN ORIGINAL UNITS	T STAT	SIG 0.01
1	Control	24.5000	24.5000		
2	5%	21.5000	21.5000	0.6557	
3	15%	21.8000	21.8000	0.5901	
4	30%	24.9000	24.9000	-0.0874	
5	60%	24.3000	24.3000	0.0437	
6	100%	24.4000	24.4000	0.0219	

Dunnett critical value = 2.9900 (1 Tailed, alpha = 0.01, df [used] = 5,40)
(Actual df = 5,54)

Dunnett's Test - TABLE 2 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	NUM OF REPS	MIN SIG DIFF (IN ORIG. UNITS)	% OF CONTROL	DIFFERENCE FROM CONTROL
1	Control	10			
2	5%	10	13.6801	55.8	3.0000
3	15%	10	13.6801	55.8	2.7000
4	30%	10	13.6801	55.8	-0.4000
5	60%	10	13.6801	55.8	0.2000
6	100%	10	13.6801	55.8	0.1000

Steel's Many-One Rank Test - Ho: Control<Treatment

GROUP	IDENTIFICATION	MEAN IN ORIGINAL UNITS	RANK SUM	CRIT. VALUE	DF	SIG 0.01
1	Control	24.5000				
2	5%	21.5000	95.50	67.00	10.00	
3	15%	21.8000	97.50	67.00	10.00	
4	30%	24.9000	105.50	67.00	10.00	
5	60%	24.3000	99.50	67.00	10.00	
6	100%	24.4000	107.50	67.00	10.00	

Critical values are 1 tailed (k = 5)

Ceriodaphnia Reproduction 210226

PMSD >47, therefore the statistics were rerun at alpha level 0.05.

ANOVA Table

SOURCE	DF	SS	MS	F
Between	5	112.7333	22.5467	0.2154
Within (Error)	54	5652.0000	104.6667	
Total	59	5764.7333		

(p-value = 0.9545)

Critical F = 3.3769 (alpha = 0.01, df = 5,54)
= 2.3861 (alpha = 0.05, df = 5,54)

Since F < Critical F FAIL TO REJECT Ho: All equal (alpha = 0.05)

Dunnett's Test - TABLE 1 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	TRANSFORMED MEAN	MEAN CALCULATED IN ORIGINAL UNITS	T STAT	SIG 0.05
1	Control	24.5000	24.5000		
2	5%	21.5000	21.5000	0.6557	
3	15%	21.8000	21.8000	0.5901	
4	30%	24.9000	24.9000	-0.0874	
5	60%	24.3000	24.3000	0.0437	
6	100%	24.4000	24.4000	0.0219	

Dunnett critical value = 2.3100 (1 Tailed, alpha = 0.05, df [used] = 5,40)
(Actual df = 5,54)

Dunnett's Test - TABLE 2 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	NUM OF REPS	MIN SIG DIFF (IN ORIG. UNITS)	% OF CONTROL	DIFFERENCE FROM CONTROL
1	Control	10			
2	5%	10	10.5689	43.1	3.0000
3	15%	10	10.5689	43.1	2.7000
4	30%	10	10.5689	43.1	-0.4000
5	60%	10	10.5689	43.1	0.2000
6	100%	10	10.5689	43.1	0.1000

Steel's Many-One Rank Test - Ho: Control<Treatment

GROUP	IDENTIFICATION	MEAN IN ORIGINAL UNITS	RANK SUM	CRIT. VALUE	DF	SIG 0.05
1	Control	24.5000				
2	5%	21.5000	95.50	75.00	10.00	
3	15%	21.8000	97.50	75.00	10.00	
4	30%	24.9000	105.50	75.00	10.00	
5	60%	24.3000	99.50	75.00	10.00	
6	100%	24.4000	107.50	75.00	10.00	

Critical values are 1 tailed (k = 5)

Title: Fathead Biomass City of Cadillac WWTP 210226
File: fb210226.txt Transform: NO TRANSFORMATION

Kolmogorov Test for Normality

D = 0.1833 (p-value = 0.0361)
D* = 0.9280

Critical D* = 1.035 (alpha = 0.01 , N = 24)
= 0.895 (alpha = 0.05 , N = 24)

Data PASS normality test (alpha = 0.01). Continue analysis.

Bartlett's Test for Homogeneity of Variance

Calculated B1 statistic = 20.7718 (p-value = 0.0009)

Data FAIL B1 homogeneity test at 0.01 level. Try another transformation.

Critical B = 15.0863 (alpha = 0.01, df = 5)
= 11.0705 (alpha = 0.05, df = 5)

Summary Statistics on Data TABLE 1 of 2

GRP	IDENTIFICATION	N	MIN	MAX	MEAN
1	Control	4	2.5500	3.4900	2.9725
2	5%	4	2.9800	3.2300	3.1225
3	15%	4	2.6500	3.6000	3.0650
4	30%	4	2.7700	3.3100	3.0700
5	60%	4	2.4000	3.4500	3.0575
6	100%	4	2.8900	6.2200	3.8425

Summary Statistics on Data TABLE 2 of 2

GRP	IDENTIFICATION	VARIANCE	SD	SEM	C.V. %
1	Control	0.1503	0.3877	0.1938	13.0420
2	5%	0.0109	0.1044	0.0522	3.3423
3	15%	0.1674	0.4091	0.2046	13.3476
4	30%	0.0504	0.2245	0.1122	7.3127
5	60%	0.2366	0.4864	0.2432	15.9075
6	100%	2.5251	1.5891	0.7945	41.3547

Fathead Biomass 210226

ANOVA Table

SOURCE	DF	SS	MS	F
Between	5	2.1007	0.4201	0.8027
Within (Error)	18	9.4218	0.5234	
Total	23	11.5225		

(p-value = 0.5623)

Critical F = 4.2479 (alpha = 0.01, df = 5,18)
= 2.7729 (alpha = 0.05, df = 5,18)

Since F < Critical F FAIL TO REJECT Ho: All equal (alpha = 0.01)

Dunnett's Test - TABLE 1 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	TRANSFORMED MEAN	MEAN CALCULATED IN ORIGINAL UNITS	T STAT	SIG 0.01
1	Control	2.9725	2.9725		
2	5%	3.1225	3.1225	-0.2932	
3	15%	3.0650	3.0650	-0.1808	
4	30%	3.0700	3.0700	-0.1906	
5	60%	3.0575	3.0575	-0.1662	
6	100%	3.8425	3.8425	-1.7006	

Dunnett critical value = 3.2100 (1 Tailed, alpha = 0.01, df = 5,18)

Dunnett's Test - TABLE 2 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	NUM OF REPS	MIN SIG DIFF (IN ORIG. UNITS)	% OF CONTROL	DIFFERENCE FROM CONTROL
1	Control	4			
2	5%	4	1.6422	55.2	-0.1500
3	15%	4	1.6422	55.2	-0.0925
4	30%	4	1.6422	55.2	-0.0975
5	60%	4	1.6422	55.2	-0.0850
6	100%	4	1.6422	55.2	-0.8700

Steel's Many-One Rank Test - Ho: Control<Treatment

GROUP	IDENTIFICATION	MEAN IN ORIGINAL UNITS	RANK SUM	CRIT. VALUE	DF	SIG 0.01
1	Control	2.9725				
2	5%	3.1225	22.00	None	4.00	
3	15%	3.0650	19.00	None	4.00	
4	30%	3.0700	20.00	None	4.00	
5	60%	3.0575	19.00	None	4.00	
6	100%	3.8425	21.00	None	4.00	

Critical values are 1 tailed (k = 5)

WARNING - There are no critical values for this combination
of groups and replicates.

PMSD >30, therefore the statistics were rerun at alpha level 0.05.

ANOVA Table

SOURCE	DF	SS	MS	F
Between	5	2.1007	0.4201	0.8027
Within (Error)	18	9.4218	0.5234	
Total	23	11.5225		

(p-value = 0.5623)

Critical F = 4.2479 (alpha = 0.01, df = 5,18)
= 2.7729 (alpha = 0.05, df = 5,18)

Since F < Critical F FAIL TO REJECT Ho: All equal (alpha = 0.05)

Dunnett's Test - TABLE 1 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	TRANSFORMED MEAN	MEAN CALCULATED IN ORIGINAL UNITS	T STAT	SIG 0.05
1	Control	2.9725	2.9725		
2	5%	3.1225	3.1225	-0.2932	
3	15%	3.0650	3.0650	-0.1808	
4	30%	3.0700	3.0700	-0.1906	
5	60%	3.0575	3.0575	-0.1662	
6	100%	3.8425	3.8425	-1.7006	

Dunnett critical value = 2.4100 (1 Tailed, alpha = 0.05, df = 5,18)

Dunnett's Test - TABLE 2 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	NUM OF REPS	MIN SIG DIFF (IN ORIG. UNITS)	% OF CONTROL	DIFFERENCE FROM CONTROL
1	Control	4			
2	5%	4	1.2329	41.5	-0.1500
3	15%	4	1.2329	41.5	-0.0925
4	30%	4	1.2329	41.5	-0.0975
5	60%	4	1.2329	41.5	-0.0850
6	100%	4	1.2329	41.5	-0.8700

Steel's Many-One Rank Test - Ho: Control<Treatment

GROUP	IDENTIFICATION	MEAN IN ORIGINAL UNITS	RANK SUM	CRIT. VALUE	DF	SIG 0.05
1	Control	2.9725				
2	5%	3.1225	22.00	10.00	4.00	
3	15%	3.0650	19.00	10.00	4.00	
4	30%	3.0700	20.00	10.00	4.00	
5	60%	3.0575	19.00	10.00	4.00	
6	100%	3.8425	21.00	10.00	4.00	

Critical values are 1 tailed (k = 5)

PARAGON Laboratories

CHAIN-OF-CUSTOD

12649 Richfield Ct. Livonia, M
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210226
CADILLAC WWT
City of Cadillac WWTP

age ____ of ____

Client Name: Cadillac WWTP	**INTERNAL USE ONLY**
Contact Person: Amy Vail	Paragon Report No.: 210226
Mailing Address: 200 N. Lake St.	Logged By: RES
City, State, Zip: Cadillac, MI 49601	Sample Condition Upon Receipt: Acceptable Other (Specify Below)
Phone and Fax: (231) 775-2368 / 775-2876	
Email: lab@cadillac-mi.net	IR Temperature Upon Receipt: 5°C
Client Job Name / No.:	Remarks: sample received chilled on natural ice - RES 8/4/16
Job Location:	
Sampled By: DS PO No.:	

WHERE WAS THE SAMPLE(s) TAKEN: Facility Does Not Chlorinate After Chlorination After Chlorination, Before Dechlorination After Dechlorination Before Chlorination

WAS THE SAMPLE(s) DECHLORINATED: YES NO **IF YES, PLEASE DESCRIBE DECHLORINATION:**

WAS THE SAMPLE(s) FILTERED: YES NO **IF YES, PLEASE STATE THE FILTER MESH SIZE:**

ANALYSIS REQUESTED

Item No.	Client Sample ID	No. of containers	Grab	No. of Grabs	**Grab Sample Period** (Start Date and Time - End Date and Time)	Comp	**Composite Sample Period** (Start Date and Time - End Date and Time)	Acute Toxicity: Ceriodaphia Dubia	Acute Toxicity: Fathead Minnows	Acute Toxicity: Daphnia Magna	Chronic Toxicity: Ceriodaphia Dubia	Chronic Toxicity: Fathead Minnows
1	Effluent	1				X	8/01/16 7:00a - 8/3/16 7:00a	X	X		X	X

Tran. #	Released By	Received By	Date	Time	Tran. #	Released By	Received By	Date	Time
1.	AMV UPS 8/3/16	R Szczygiel	8/4/15	10:20	3.				
2.					4.				

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CHAIN-OF-CUSTODY RECORD

Client Name: Cadillac WWTP	**INTERNAL USE ONLY**
Contact Person: Amy Vail	Paragon Report No.: 216226
Mailing Address: 300 N. Lake St	Logged By: RES
City, State, Zip: Cadillac, MI 49601	Sample Condition Upon Receipt: Acceptable Other (Specify Below)
Phone and Fax: (231) 775-3368 / 775-7826	
Email: lab@cadillac-mi.net	IR Temperature Upon Receipt: 4°C
Client Job Name / No.:	Remarks: samples received chilled on natural ice - RES 8/5/16
Job Location:	
Sampled By: DS PO No.:	

WHERE WAS THE SAMPLE(s) TAKEN: Facility Does Not Chlorinate After Chlorination After Chlorination, Before Dechlorination After Dechlorination Before Chlorination

WAS THE SAMPLE(s) DECHLORINATED: YES NO **IF YES, PLEASE DESCRIBE DECHLORINATION:**

WAS THE SAMPLE(s) FILTERED: YES NO **IF YES, PLEASE STATE THE FILTER MESH SIZE:**

ANALYSIS REQUESTED

Item No.	Client Sample ID	No. of containers	Grab	No. of Grabs	Grab Sample Period (Start Date and Time - End Date and Time)	Comp	Composite Sample Period (Start Date and Time - End Date and Time)	Acute Toxicity: Ceriodaphia Dubia	Acute Toxicity: Fathead Minnows	Acute Toxicity: Daphnia Magna	Chronic Toxicity: Ceriodaphia Dubia	Chronic Toxicity: Fathead Minnows
1	Effluent	1				X	8/13/16 710 am - 8/14/16 710 am	✓	✓		X	✓

Tran. #	Released By	Received By	Date	Time	Tran. #	Released By	Received By	Date	Time
1.	[illegible] 3/15	AIS	8/5/16	10:05	3.				
2.					4.				

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CHAIN-OF-CUSTODY RECORD

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Page ____ of ____

Client Name: Cadillac WWTP	**INTERNAL USE ONLY**
Contact Person: Amy Vail	Paragon Report No.: 210226
Mailing Address: 300 N. Lake St.	Logged By: RES
City, State, Zip: Cadillac, MI 49601	Sample Condition Upon Receipt: Acceptable Other (Specify Below)
Phone and Fax: (231) 775-2368 / 775-2876	
Email: lab@cadillac-mi.net	IR Temperature Upon Receipt: 5°C
Client Job Name / No.:	Remarks: sample received chilled on natural ice RES 8-9-16
Job Location:	
Sampled By: DS. PO No.:	

WHERE WAS THE SAMPLE(s) TAKEN: Facility Does Not Chlorinate After Chlorination After Chlorination, Before Dechlorination After Dechlorination Before Chlorination

WAS THE SAMPLE(s) DECHLORINATED: YES NO **IF YES, PLEASE DESCRIBE DECHLORINATION:**

WAS THE SAMPLE(s) FILTERED: YES NO **IF YES, PLEASE STATE THE FILTER MESH SIZE:**

ANALYSIS REQUESTED

Item No.	Client Sample ID	No. of containers	Grab	No. of Grabs	**Grab Sample Period** (Start Date and Time - End Date and Time)	Comp	**Composite Sample Period** (Start Date and Time - End Date and Time)	Acute Toxicity: Ceriodaphia Dubia	Acute Toxicity: Fathead Minnows	Acute Toxicity: Daphnia Magna	Chronic Toxicity: Ceriodaphia Dubia	Chronic Toxicity: Fathead Minnows
1	Effluent	1				X	8/7/16 7:00am - 8/8/16 7:00a	X	X		X	X

Tran. #	Released By	Received By	Date	Time	Tran. #	Released By	Received By	Date	Time
1.	AV UPS 3:15p	ECP	8/9/16	10:30	3.				
2.					4.				